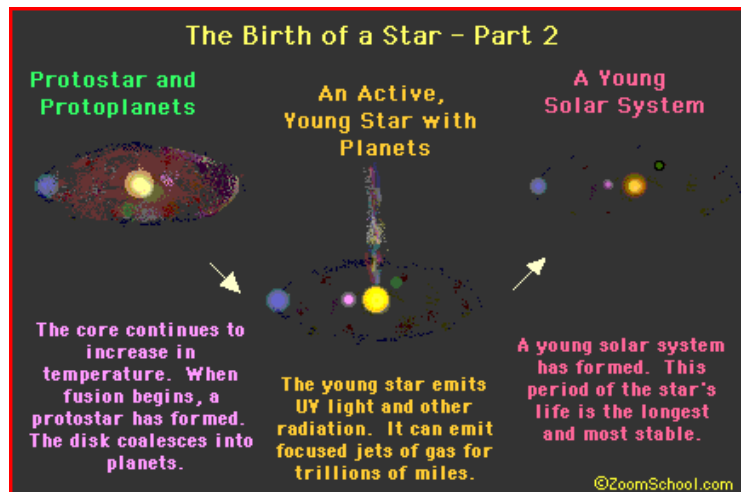
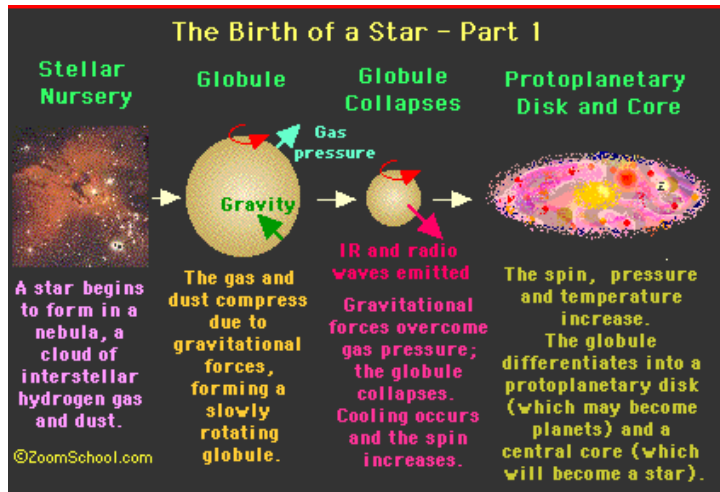


LIFE CYCLES OF STARS

Birth of Stars

Astronomers believe that stars are born from the gas and dust particles in a **nebula**. Gravity starts to pull the particles together until the temperature rises, forming a **protostar**. If the temperature raises high enough, **nuclear fusion** starts and the young star begins to glow.

In the fusion process, hydrogen gas is converted into helium gas, releasing huge amounts of energy as matter is annihilated and turned into pure energy according to Einstein's equation ($E = mc^2$).



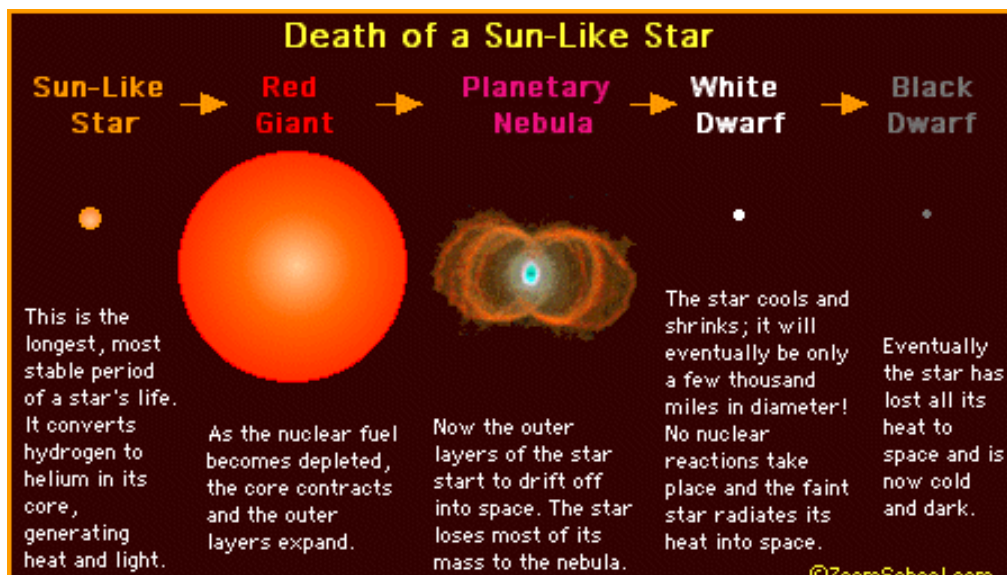
Size

The mass of a star determines its life cycle.

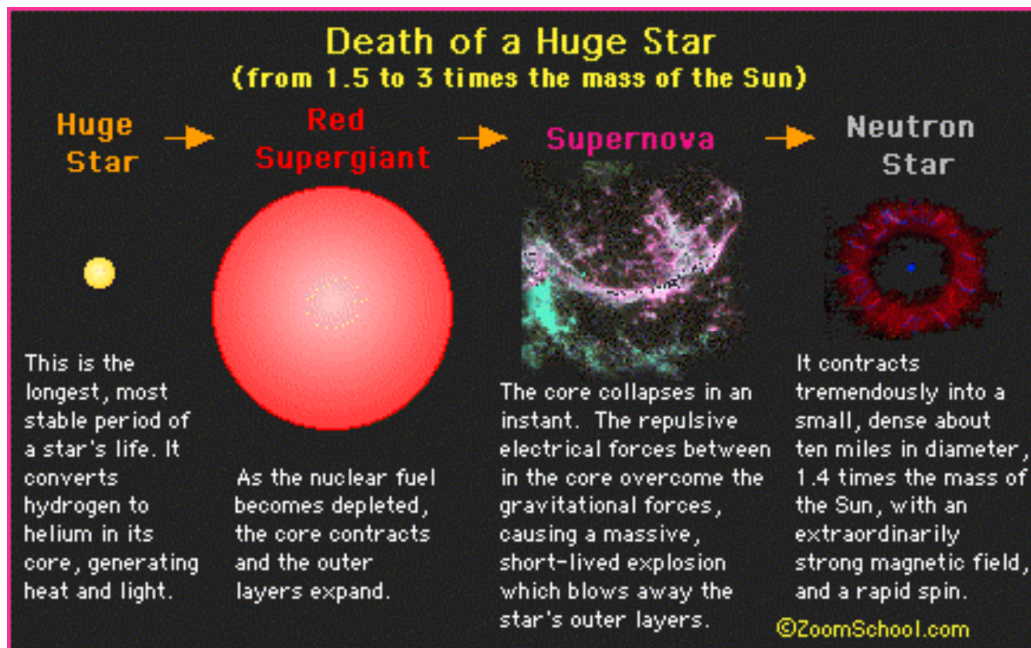
Protostar < 0.1 x mass of our Sun - shrinks but does not get hot enough for fusion to start. Forms a red dwarf before dying out.

There are three possible lifecycles for stars.

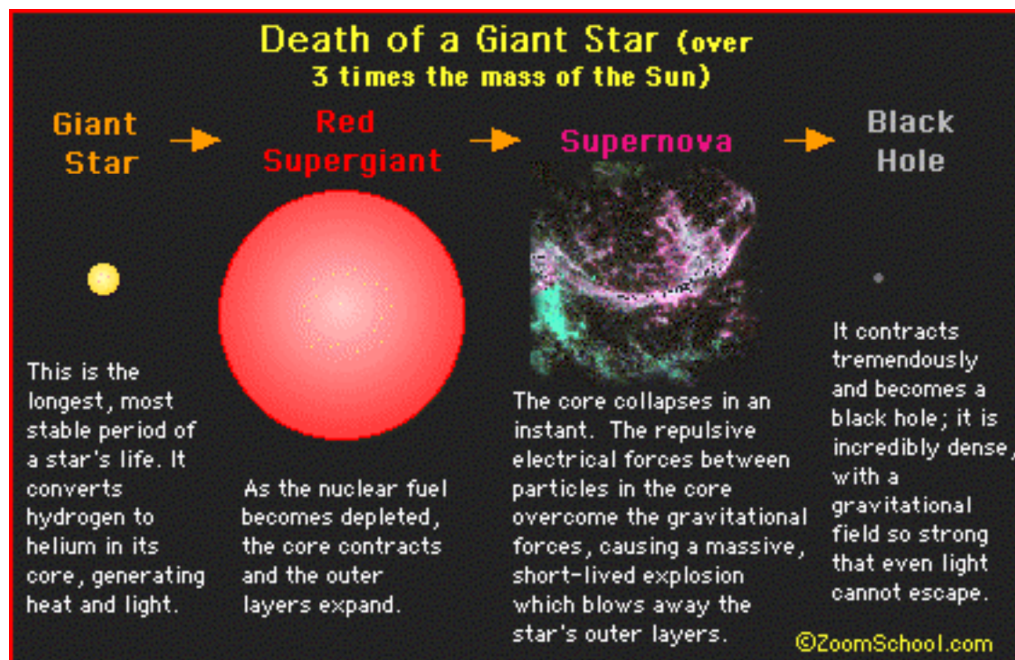
Stars up to 1.5 x mass of Sun - ultimately becomes a **black dwarf**.



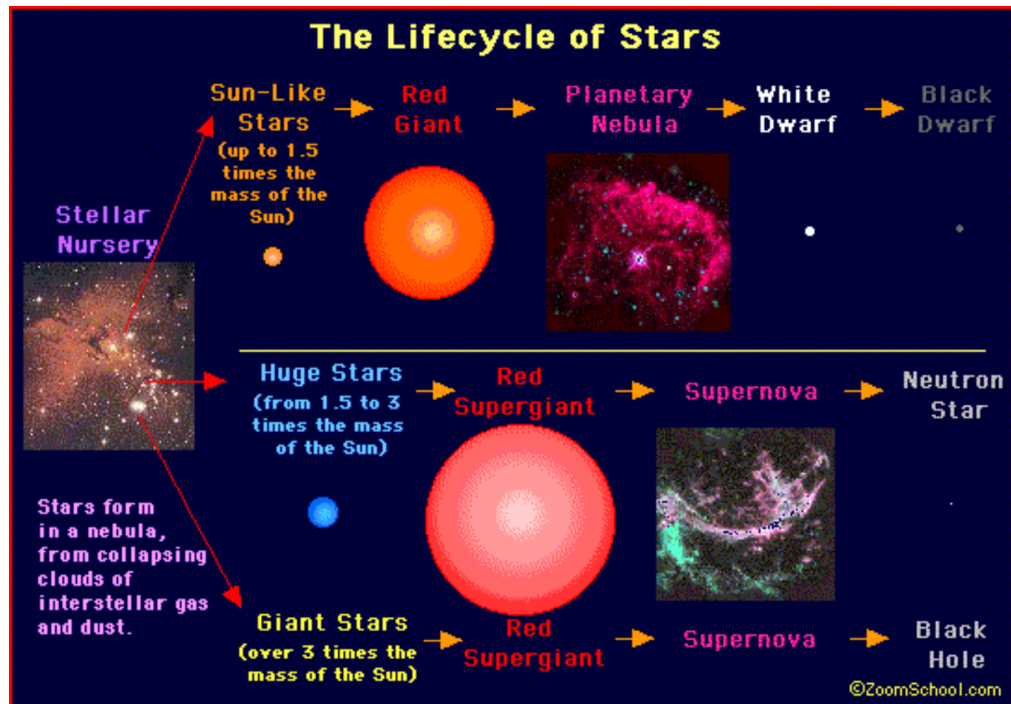
Stars 1.5 - 3.0 x mass of Sun - becomes a *neutron star*.



Stars > 3.0 x mass of Sun - becomes a black hole.



Life Cycle Summary



Hertzsprung-Russell Diagram

This diagram plots the absolute magnitude of brightness against the temperature of a star.

Stars that are on the **main sequence** are stable and live for billions of years. Our Sun is halfway through its life and has 5 billion years left.

As stars age and die, they evolve and move from the main sequence towards the top of the diagram. After a nova or supernova explosion, stars move down towards the lower left of the diagram.

